Homework 7

This homework is due at 11 PM on March 13, 2026.

Submission Format: Your homework submission should consist of a single PDF file that contains all of your answers (any handwritten answers should be scanned).

1. **Convex or Concave**

   Determine whether the following functions are convex, strictly convex, concave, strictly concave, both or neither.

   (a) $f(x) = \mathrm{e}^x - 1$ on $\mathbb{R}$.

   (b) $f(x_1, x_2) = x_1 x_2$ on $\mathbb{R}^2_{++}$, i.e., when $x_1 > 0$ and $x_2 > 0$.

   (c) The log-likelihood of a set of points $\{x_1, \ldots, x_n\}$ that are normally distributed with mean μ and finite variance $\sigma > 0$ is given by:

   $$f(\mu, \sigma) = n \log\left(\frac{1}{\sqrt{2\pi}\sigma}\right) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2 \tag{1}$$

   i. Show that if we view the log likelihood for fixed σ as a function of the mean, i.e

   $$g(\mu) = n \log\left(\frac{1}{\sqrt{2\pi}\sigma}\right) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2 \tag{2}$$

   then g is strictly concave (equivalently, we say f is strictly concave in μ).

   ii. Show that if we view the log likelihood for fixed μ as a function of the inverse of the variance, i.e

   $$h(z) = n \log\left(\frac{\sqrt{z}}{\sqrt{2\pi}}\right) - \frac{z}{2}\sum_{i=1}^{n}(x_i - \mu)^2 \tag{3}$$

   then h is strictly concave (equivalently, we say f is strictly concave in $z = \frac{1}{\sigma^2}$). Note that we have used the dummy variable z to denote $\frac{1}{\sigma^2}$.

   iii. Show that f is not jointly concave in $\mu, \frac{1}{\sigma^2}$. *HINT: We say a function $w(\vec{x}, \vec{y})$ with $\vec{x} \in \mathbb{R}^m$ and $\vec{y} \in \mathbb{R}^n$ is jointly convex if*

   $$w\left(\lambda(\vec{x}_1, \vec{y}_1) + (1-\lambda)(\vec{x}_2, \vec{y}_2)\right) \leq \lambda w((\vec{x}_1, \vec{y}_1)) + (1-\lambda) w((\vec{x}_2, \vec{y}_2)). \tag{4}$$

   This is the same as letting $\vec{z} = (\vec{x}, \vec{y})$ and saying f is convex in $\vec{z}$. We can define joint concavity in a similar fashion by reversing the inequalities.

   (d) $f(x) = \log(1 + \mathrm{e}^x)$. Note that this implies that $g(x) = -f(x) = \log\left(\frac{1}{1+\mathrm{e}^x}\right)$ is concave. Compare this to $h(x) = \frac{1}{1+\mathrm{e}^x}$, is $h(x)$ convex or concave?

2. Convex and strictly convex functions

(a) Recall that a function $f : \mathbb{R}^n \to \mathbb{R}$ is said to be strictly convex if it satisfies Jensen's inequality with strict inequality, i.e., $\forall \vec{x} \neq \vec{y} \in \mathbb{R}^n$ and $\forall t \in (0, 1)$, we have

$$f(t\vec{x} + (1-t)\vec{y}) < tf(\vec{x}) + (1-t)f(\vec{y})$$

Show that for a strictly convex function $f : \mathbb{R}^n \to \mathbb{R}$, the problem

$$\min_{\vec{x} \in \mathbb{R}^n} f(\vec{x}) \tag{5}$$

has at most one solution.

HINT: Try to argue by contradiction assuming that there are two solutions $\vec{x}_1, \vec{x}_2$ which achieve the minimum value. Argue that using these two points you can find another point in $\mathbb{R}^n$ with strictly smaller function value.

(b) Prove that for all convex optimization problems $\min_{\vec{x} \in \mathcal{X}} f(\vec{x})$, where f is a convex function and $\mathcal{X}$ is a convex set, all local minima are global minima. You may not assume that f is differentiable.

HINT: Start with assuming $\vec{x}^\star$ is a local minimum that is not global, and $\vec{\tilde{x}}$ is a global minimum. Use the definition of the convexity of a function to prove by contradiction.

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3. Quadratic inequalities

Consider the set S defined by the following inequalities:

$$(x_1 \geq -x_2 + 1 \text{ and } x_1 \leq 0) \text{ or } (x_1 \leq -x_2 + 1 \text{ and } x_1 \geq 0)\,. \tag{6}$$

To be more precise,

$$S_1 = \{\vec{x} \in \mathbb{R}^2 \mid x_1 \geq -x_2 + 1, x_1 \leq 0\} \tag{7}$$

$$S_2 = \{\vec{x} \in \mathbb{R}^2 \mid x_1 \leq -x_2 + 1, x_1 \geq 0\} \tag{8}$$

$$S = S_1 \cup S_2. \tag{9}$$

(a) Draw the set S. Is it convex?

(b) Show that the set S, can be described as a single quadratic inequality of the form $q(\vec{x}) = \vec{x}^\top A\vec{x} + 2\vec{b}^\top \vec{x} + c \leq 0$, for matrix $A = A^\top \in \mathbb{R}^{2\times 2}$, $\vec{b} \in \mathbb{R}^2$ and $c \in \mathbb{R}$ i.e S can be written as $S = \{\vec{x} \in \mathbb{R}^2 \mid q(\vec{x}) \leq 0\}$). Find $A, \vec{b}, c$.
Hint: Can you combine the constraints to make one quadratic constraint?

(c) The convex hull of a set $A \subseteq \mathbb{R}^n$ is the set of all convex combinations of points in A, i.e.,

$$\operatorname{conv}(A) = \left\{ \sum_{i=1}^{k} \theta_i \vec{x}_i \;\middle|\; k \in \mathbb{N}, \theta_1, \ldots, \theta_k \geq 0, \sum_{i=1}^{k} \theta_i = 1, \vec{x}_1, \ldots, \vec{x}_k \in A \right\}. \tag{10}$$

What is the convex hull of S?

(d) We will now consider some convex optimization problems over S_1 that illustrate the role of the constraints in the optimization problem. For each of the following optimization problems find the optimal point, $\vec{x}^*$. Describe the constraints that are active in attaining the optimal value. *Hint: Suppose that there exists a point $\vec{x}$ such that $\nabla f(\vec{x}) = 0$. From the first order characterization of a convex function $\vec{x}$ would be an optimum value for f subject to no constraints. If $\vec{x}$ is not in the constraint set S_1, then the optimum point must be on the boundary of the set, i.e. it satisfies at least one of the constraints defining S_1 with equality.*

i. Minimize $f(\vec{x}) = (x_1 + 1)^2 + (x_2 - 3)^2$ subject to $\vec{x} \in S_1$.

ii. Minimize $f(\vec{x}) = (x_1 + 2)^2 + (x_2 - 2)^2$ subject to $\vec{x} \in S_1$.

iii. Minimize $f(\vec{x}) = x_1^2 + x_2^2$ subject to $\vec{x} \in S_1$.

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4. Quadratics and Least Squares

In this question, we will see that every least squares problem can be considered as minimization of a quadratic cost function; whereas not every quadratic minimization problem corresponds to a least-squares problem. To begin with, consider the quadratic function, $f : \mathbb{R}^2 \to \mathbb{R}$ given by:

$$f(\vec{w}) = \vec{w}^\top A\vec{w} - 2\vec{b}^\top \vec{w} + c \tag{11}$$

where $A \in \mathbb{S}^2_+$ (set of symmetric positive semidefinite matrices in $\mathbb{R}^{2\times 2}$), $\vec{b} \in \mathbb{R}^2$ and $c \in \mathbb{R}$.

(a) Assume $c = 0$, and assume that setting $\nabla f(\vec{w}) = 0$ allows us to find the unique minimizer. Give a concrete example of a matrix $A \succ 0$ and a vector $\vec{b}$ such that the point $\vec{w}^\star = \begin{bmatrix} -1 & 1 \end{bmatrix}^\top$ is the unique minimizer of the quadratic function $f(\vec{w})$.

(b) Assume $c = 0$. Give a concrete example of a matrix $A \succeq 0$, and a vector $\vec{b}$ such that the quadratic function $f(\vec{w})$ has infinitely many minimizers and all of them lie on the line $w_1 + w_2 = 0$.

HINT: Take the gradient of the expression and set it to zero. What needs to be true for there to be infinitely many solutions to the equation?

(c) Assume $c = 0$. Let $\vec{w} = \begin{bmatrix} 1 & 0 \end{bmatrix}^\top$. Give a concrete example of a **non-zero** matrix $A \succeq 0$ and a vector $\vec{b}$ such that the quadratic function $f(\alpha\vec{w})$ tends to $-\infty$ as $\alpha \to \infty$. *HINT: Use the eigenvalue decomposition to write $A = \sigma_1\vec{u}_1\vec{u}_1^\top + \sigma_2\vec{u}_2\vec{u}_2^\top$ and express $\vec{w}$ in the basis formed by $\vec{u}_1, \vec{u}_2$.*

(d) Say that we have the data set $\{(\vec{x}_i, y_i)\}_{i=1,\ldots,n}$ of data points $\vec{x}_i \in \mathbb{R}^d$ and values $y_i \in \mathbb{R}$. Define $X = \begin{bmatrix} \vec{x}_1 & \ldots & \vec{x}_n \end{bmatrix}^\top$ and $\vec{y} = \begin{bmatrix} y_1 & \ldots & y_n \end{bmatrix}^\top$. In terms of X and $\vec{y}$, find a matrix A, a vector $\vec{b} \in \mathbb{R}^d$ and a scalar c, so that we can express the sum of the square losses $\sum_{i=1}^{n}(\vec{w}^\top\vec{x}_i - y_i)^2$ as the quadratic function $f(\vec{w}) = \vec{w}^\top A\vec{w} - 2\vec{b}^\top\vec{w} + c$.

(e) Here are three statements with regards to the minimization of a quadratic loss function:

i. It can have a unique minimizer.

ii. It can have infinitely many minimizers.

iii. It can be unbounded from below, i.e. there is some direction, $\vec{w}$ so that $f(\alpha\vec{w})$ goes to $-\infty$ as $\alpha \to \infty$.

All three statements apply to general minimization of a quadratic cost function. Parts (a), (b) and (c) give concrete examples of quadratic cost functions where (i), (ii) and (iii) apply respectively. However, notice that statement (iii) cannot apply to the least squares problem as the objective is always positive. The least-squares problem can have infinitely many minimizers though. How? Consider the gradient of the least squares problem in part (d) at an optimal solution $\vec{w}^\star$:

$$\nabla f(\vec{w}^\star) = 2X^\top X\vec{w}^\star - 2\vec{b} = 0. \tag{12}$$

Therefore, the least squares problem only has multiple solutions if $X^\top X$ is not full rank. This means that $\operatorname{rank}\left(X^\top X\right) = \operatorname{rank}(X) < d$. Finally, the rank of X is less than d when the data points $\{\vec{x}_i\}_{i=1}^n$ do not span $\mathbb{R}^d$. This can happen when the number of data points n is less than d or when $\{\vec{f}_i\}_{i=1}^d$ are linearly dependent where $\vec{f}_i$ are the columns of X, i.e., the features.

We will see soon that these cases correspond to the *convexity* of the function: if the function is strictly convex, then it has a unique minimizer; and if it is just convex, then it can have multiple minimizers; and in

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both cases, it can have no minimizers. We will see soon how to prove that the quadratic objective functions we discuss in this problem are convex, strictly convex, or even non-convex.

Indicate below that you have read and understood the discussion above.

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5. Direction of Steepest Ascent

For a differentiable function $f : \mathbb{R}^n \mapsto \mathbb{R}$ we want to show that the gradient $\nabla f(\vec{x})$ is the direction of steepest ascent at the point $\vec{x}$.

(a) Let us define the rate of change of the function $f(\vec{x})$ at the point $\vec{x}$ along an arbitrary unit vector $\vec{u}$ as:

$$D_{\vec{u}} f(\vec{x}) = \lim_{h \to 0} \frac{f(\vec{x} + h\vec{u}) - f(\vec{x})}{h}. \tag{13}$$

We call this the directional derivative. Show that the directional derivative can be equivalently expressed as $D_{\vec{u}} f(\vec{x}) = \vec{u}^\top [\nabla f(\vec{x})]$.

HINT: Use Taylor approximation of the function around the point $\vec{x}$ and evaluate it at the point $\vec{x} + h\vec{u}$.

(b) Show that

$$\frac{\nabla f(\vec{x})}{\|\nabla f(\vec{x})\|_2} = \operatorname*{argmax}_{\|\vec{u}\|_2 = 1} \vec{u}^\top [\nabla f(\vec{x})]. \tag{14}$$

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6. Gradient Descent Algorithm

Given a continuous and differentiable function $f : \mathbb{R}^n \to \mathbb{R}$, the gradient of f at any point $\vec{x}$, $\nabla f(\vec{x})$, is orthogonal to the level curve of f at point $\vec{x}$, and it points in the increasing direction of f. In other words, moving from point $\vec{x}$ in the direction $\nabla f(\vec{x})$ leads to an increase in the value of f, while moving in the direction of $-\nabla f(\vec{x})$ decreases the value of f. This idea gives an iterative algorithm to minimize the function f: the gradient descent algorithm.

(a) Consider $f(x) = \frac{1}{2}(x-2)^2$, and assume that we use the gradient descent algorithm:

$$x_{k+1} = x_k - \eta \nabla f(x_k) \quad \forall k \geq 0, \tag{15}$$

with some random initialization x_0, where $\eta > 0$ is the step size (or the learning rate) of the algorithm. Write $(x_k - 2)$ in terms of $(x_0 - 2)$, and show that x_k converges to 2, which is the unique minimizer of f, when $\eta = 0.2$.

(b) Let $\alpha, \beta \in \mathbb{R}$ and that for all η such that $\alpha < \eta < \beta$, the gradient descent algorithm converges to 2 from all possible initializations in $\mathbb{R}$. What are the smallest α and the largest β? After you determine α, β, answer the following. What happens when we set (i) $\eta = \alpha$ (ii) $\eta = \beta$ (iii) $\eta > \beta$?

(c) Now assume that we use the gradient descent algorithm to minimize $f(\vec{x}) = \frac{1}{2}\left\|A\vec{x} - \vec{b}\right\|_2^2$ for some $A \in \mathbb{R}^{m \times n}$ and $\vec{b} \in \mathbb{R}^m$, where A has full column rank. First compute $\nabla f(\vec{x})$. Note that $(A^\top A)^{-1}A^\top \vec{b}$ is the solution to the least-squares problem, and $(\vec{x}_k - (A^\top A)^{-1}A^\top \vec{b})$ is the distance from the solution at time k. Write $(\vec{x}_k - (A^\top A)^{-1}A^\top \vec{b})$ in terms of $(\vec{x}_0 - (A^\top A)^{-1}A^\top b)$.

(d) Now consider $f(\vec{x}) = \frac{1}{2}\left\|A\vec{x} - \vec{b}\right\|_2^2 + \frac{1}{2}\lambda \left\|\vec{x}\right\|_2^2$ for some $A \in \mathbb{R}^{m \times n}$, $\vec{b} \in \mathbb{R}^m$, and $\lambda \in \mathbb{R}$, where A has full column rank and $\lambda \geq 0$. Suppose we solve this problem via gradient descent with step-size $\eta = \frac{1}{\sigma_1^2 + \lambda}$, where σ_1 is the maximum singular value of A. Show the gradient descent converges.

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7. Homework Process

With whom did you work on this homework? List the names and SIDs of your group members.

NOTE: If you didn't work with anyone, you can put "none" as your answer.

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